

# Alan Nguyen Pham

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**Mechanical R&D | Soft Robotics | Physical Interaction**

Mechanical engineering Ph.D. candidate building soft robotic materials, reconfigurable pneumatic mechanisms, 3D-printed robotic cells, and mechanism simulators for human-facing physical systems. Targeting mechanical R&D roles where CAD/fabrication, experimental characterization, pneumatic/compliant mechanisms, and visible physical behavior meet.

## Research Focus

Soft robotics; reconfigurable mechanisms; robotic materials; pneumatic soft actuation; Sarrus/PneuNet robotic cells; morphing surfaces; shape displays; compliant mechanisms; mechanism simulation; inverse design; CAD/fabrication; experimental characterization; force/stiffness testing; haptics; physical HCI; adaptive interfaces.

## Education

### Doctor of Philosophy, Mechanical Engineering

Worcester Polytechnic Institute, Worcester, MA

**Expected 2027**

Advisors: Dr. Pratap Rao and Dr. Cagdas Onal

### Master of Science, Mechanical Engineering

California State University, Long Beach, Long Beach, CA

**2022**

Advisor: Dr. Allen Teagle-Hernandez

### Bachelor of Science, Mechanical Engineering

California State University, Maritime Academy, Vallejo, CA

**2019**

Advisor: Dr. Tomas Oppenheim

## Awards

NSF NRT Future of Robots in the Workplace – Research and Development (FORW-RD) Fellowship

2024

WPI Intercollegiate Soft Robotics Competition – 1st Place

2024

## Research Experience

### Worcester Polytechnic Institute, Worcester, MA

*Graduate Researcher – NanoEnergy Lab and Soft Robotics Lab*

**Aug 2023 – Present**

*Advisors: Dr. Pratap Rao and Dr. Cagdas Onal*

- Created a monolithically printed Sarrus/PneuNet soft-robotic architecture that reconfigures from planar morphing sheets into cylindrical continuum robots and crawling/rolling locomotion assemblies.
- Demonstrated programmable behaviors: single- and double-dome shape displays, bilayer overhangs, tennis-ball transport/capture, continuum bending/grasping, 200 mm peristaltic crawling, self-rolling, and horizontal, diagonal, and circular traveling waves.
- Characterized pressure-dependent expansion, bending angle, force-displacement response, hysteresis, and variable stiffness to link geometry and actuation patterns to macroscopic behavior.
- Built Wavevis, a public React/TypeScript/Three.js mechanism visualizer for programming double-layer Sarrus-cell arrays, inspecting inverse sheet deformation, and exporting geometry/metrics for mechanism design.
- Developing FluxCell, a 3D-printed pneumatic-mechanical robotic-cell system with inter-cell connectors that physically join neighboring cells, pass shared manifold pressure, and route air through magnet- or electropermanent-magnet valve concepts into individual PneuNets.

### NASA L' SPACE (NPWEE), Tempe, AZ

*Soft Robotics Graduate Researcher – Project SELENE*

**Aug 2025 – Present**

- Advanced Project SELENE, a multidisciplinary exploration-robotics effort developing adaptable, resilient robotic systems for extreme and remote environments, including beyond-Earth mission concepts.
- Developed a modular cable-driven soft robotic arm concept for terrain navigation and regolith sample collection in hard-to-reach exploration settings.
- Refined the “Spirobs” arm design by integrating triple-cable actuation with an elastic braided-steel backbone for multidirectional bending, mechanical compliance, dust-resistant operation, and reduced maintenance.
- Led CAD updates, fabrication planning, 3D-printing decisions, assembly strategy, and structured test/evaluation planning while guiding undergraduate teammates and coordinating with NASA subject matter experts.

**Sandia National Laboratories, Albuquerque, NM****June 2025 – Aug 2025***Graduate Research Intern – Nonlinear Mechanics and Dynamics (NOMAD) Research Institute*

- Co-developed a NOMAD parameter study quantifying how finite-element setup choices affect nonlinear-dynamics predictions for an electromechanical ratcheting mechanism with intermittent contact, friction, springs, and pawl-gear interactions.
- Built comparable analytical, MATLAB multibody, and finite-element submodels for pin-spring-pawl, pin-pawl, and pawl-gear interactions under haversine shock and sinusoidal vibration.
- Ran structured sweeps across momentum-balance iterations (1–100), mesh density, and processor count (50–250%) to separate physical mechanism response from solver and modeling sensitivity.
- Identified sensitivity regimes in which pin-spring-pawl models converged while pin-pawl and pawl-gear cases showed contact/friction divergence, mesh-related lock-up, and environment-dependent sensitivity.
- Presented final results showing that simulation sensitivity appears at submodel scale, informing more defensible setup choices before full assembly-level ratcheting studies.

**Worcester Polytechnic Institute, Worcester, MA****Aug 2023 – Present***NSF NRT FORW-RD Fellow and Trainee*

- Selected as an NSF NRT FORW-RD trainee in an interdisciplinary program on robotic interfaces and assistants for future workplaces spanning robotics, mechanical engineering, computer science, materials, business, humanities, and social sciences.
- Connected doctoral research to haptic and virtual training, soft robotic teleoperation, compliant interaction, and human guidance of robots for high-skill physical tasks.
- Developed broader-impact framing for robotics and AI systems that operate near people, emphasizing workplace adoption, trust, physical interaction, training, and responsible deployment.
- Secured a \$1,000 grant supporting equipment, travel, and professional-development costs.

**Worcester Polytechnic Institute, Worcester, MA****Aug 2023 – Dec 2023***Research Assistant – Cognitive Medical Technology Laboratory*

- Fabricated and assembled low-cost open-source actuation hardware for concentric tube continuum robots, a miniaturized robot class for precise navigation in constrained spaces and minimally invasive surgery.
- Contributed to an accessible CTR research platform with easy-to-source components, bills of materials, assembly instructions, and guided learning materials for early graduate students.
- Integrated actuation hardware, MATLAB forward kinematics, and optical-tracking experiments to validate end-effector displacement.
- Coordinated sourcing, fabrication, and assembly across WPI's Cognitive Medical Technology Lab, PracticePoint, and Innovation Studio for two complete CTR systems.

**California State University, Long Beach, Long Beach, CA****Sept 2022 – Aug 2023***Research Assistant – Multiscale Mechanics of Materials Group**Advisor: Dr. Mortaza Saeidi-Javash*

- Built a literature foundation for a flexible-electronics research program covering printed and conformal sensors, 2D nanomaterial inks, and wearable human-machine-interface devices.
- Synthesized fabrication approaches across inkjet, screen, aerosol-jet, extrusion, and direct-ink-write printing using graphene, MXene, MoS<sub>2</sub>, thermoelectric, and piezoelectric material systems.
- Evaluated sensitivity, selectivity, response time, mechanical compliance, and durability of flexible physical, chemical, and biosensors for soft robotics, human-robot interfaces, structural health monitoring, and wearable systems.
- Collaborated with graduate researchers on a direct-ink-write temperature sensor printed on a flexible PET substrate.

**California State University, Long Beach, Long Beach, CA****April 2021 – July 2022***Graduate Researcher – Violin Acoustics and Structural Dynamics**Advisor: Dr. Allen Teagle-Hernandez*

- Completed a master's thesis translating perceived violin quality into measurable vibroacoustic features, linking structural design, harmonic balance, frequency response, and performer/listener perception.
- Designed a controlled comparison of high- and low-cost modern violins, standardizing strings and fittings while documenting top plate, bridge, f-hole, bass bar, sound post, and tailpiece variables.
- Built a LabVIEW/NI data-acquisition workflow with PCB accelerometers, an array microphone, and an impulse modal hammer; collected 67-point top-plate maps and anechoic-chamber acoustic measurements.
- Linked FFT-derived violin response features to modal balance and perceived tone.

**California State University, Maritime Academy, Vallejo, CA***Undergraduate Researcher – Cross-Flow Tidal Turbine***Sept 2018 – May 2019***Advisor: Dr. Tomas Oppenheim*

- Conceptualized a Darrieus cross-flow tidal turbine, defined rotor geometry and build constraints, and selected 3D-printed blades with fiberglass coating as the manufacturing approach.
- Verified performance through water-tunnel power assessment and ANSYS structural analysis, evaluating blade and hub stresses under hydrodynamic loading and documenting operation around 25 rpm.
- Fabricated a full-scale prototype using additive manufacturing and sheet-metal forming informed by scaled water-tunnel experiments.

**Publications and Research Outputs**

- Pham, A. N., Rao, P., & Onal, C. D. (2026). Manuscript in prep.: *Reconfigurable Sarrus-linkage cells for multifunctional soft robots*.
- Pham, A. N. (2026). *Wavevis: interactive 3D visualization for double-layer Sarrus-cell arrays and inverse sheet deformation* [Public simulator and mechanism-design tool].
- Pham, A. N. (2026). *FluxCell pneumatic-mechanical robotic-cell system* [Research platform for 3D-printed inter-cell connectors, shared manifold pressure, and magnet- or electropermanent-magnet valve concepts for PneuNet actuation].
- Winston, S., Pham, A., Bahr, B., Singh, A., Schumann, C., Flicek, R., Grutzik, S., Kuether, R. J., & Vemaganti, K. (2025). *Quantifying the Effect of Non-Physical Parameters on the Nonlinear Dynamics of an Electromechanical Ratcheting Mechanism*. Accepted to IMAC, A Conference and Exposition on Structural Dynamics.
- Bonofiglio, K. (2023). *Making Concentric Tube Robots More Accessible: An Open-Source Platform with Easy-to-Source Components and Guided Learning Materials* [Master's thesis, Worcester Polytechnic Institute]. Research contributor to open-source CTR hardware assembly and validation support.
- Pham, A., & Teagle Hernandez, A. (2022). *Structural Properties and Perceived Acoustic Quality of Modern Violins* (Publication No. 29065875) [Master's thesis, California State University, Long Beach]. ProQuest Dissertations and Theses Global.

**Professional Experience****Worcester Polytechnic Institute, Worcester, MA***Teaching Assistant***Aug 2023 – Present**

- Developed learning activities for Stress Analysis, Statics, and Computer Aided Design with professors and teaching teams.
- Led weekly conference sessions and office hours for 90+ student groups, translating difficult engineering material into interactive problem-solving sessions.
- Designed hands-on learning sessions and guided students in connecting classroom mechanics concepts to physical models and engineering judgment.

**NLS Lighting, Carson, CA***Mechanical Engineer***April 2022 – July 2023**

- Led design-to-release luminaire product improvements across thermal testing, IP testing, FEA static simulation, fabrication, assembly, and installation support.
- Produced complete product design packages, including 2D drawings, bills of materials, engineering change orders, installation instructions, and application documentation.
- Collaborated with Washington, DC stakeholders to develop a retrofit kit for upgrading more than 75,000 street and alley lights to energy-efficient LEDs as part of a multi-million-dollar Smart Street Lighting project.

**Hubbell, City of Industry, CA***Mechanical Engineer Intern***May 2021 – Aug 2021**

- Managed fixture assembly and DLC V5.1 certification planning, reducing test cost and labor while accounting for SKU quantities and sales volume.
- Performed impact testing and FEA validation on luminaire assemblies, checking bracket, lens, and LED-PCB designs in Solid Edge.
- Supported launch of the Prevent vandal-resistant luminaire series through baseline testing, part sourcing, and assembly coordination with new product design teams.

**Academy Inc., Montebello, CA****July 2019 – Aug 2020***Mechanical Design Engineer*

- Implemented parametric product and pricing configurations, reducing design and quoting time.
- Engineered shade sails, awnings, and tensioned fabric structures for manufacturability and field installation.
- Managed projects from site measurements and material specifications through coordination and scope execution.

**American Metal Bearing Company, Garden Grove, CA****May 2019 – June 2019***Mechanical Engineer Intern*

- Optimized thrust, line-shaft, stern-tube, and strut bearing components using SolidWorks/Abaqus FEA, reducing material and labor cost.
- Coordinated manufacturing flow for line-shaft spherical bearings from rough casting through inspection, finishing, packaging, and shipment.
- Revised bearing-component drawings under Engineering Change Requests while maintaining GD&T standards.

**Technical Skills**

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| <b>Mechanical design</b> | SolidWorks, Onshape, PTC Creo, FEA review, GD&T-aware drawings, BOMs, ECOs, DFM, assembly planning, fixture design, prototype iteration.                                              |
| <b>Soft robotics</b>     | Pneumatic actuation, compliant mechanisms, continuum robots, modular soft robots, morphing surfaces, shape displays, haptics, human-robot interaction, robotic prototyping.           |
| <b>Software</b>          | CUBIT, Sierra/SolidMechanics, ParaView, MATLAB, Simulink Simscape Multibody, Abaqus, Ansys FEA, Python, C/C++, TypeScript, React, Three.js, ROS2, LabVIEW, Minitab, MS Office.        |
| <b>Experiments</b>       | Modal testing, vibroacoustic measurement, FFT-based signal analysis, anechoic-chamber testing, optical tracking, pressure/force/displacement characterization, design of experiments. |
| <b>Hardware</b>          | 3D printing, laser cutting, soldering, Arduino, Raspberry Pi, National Instruments DAQ, strain gages, thermocouples, pressure sensors, force testing (Mecmesin), hand tools.          |

**Relevant Coursework**

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Soft Robotics; Advanced Dynamics; Ethics and Communication in Robotics and AI Research; Foundations in Robotics; Advanced Mechanics of Deformable Bodies; Mechatronic System Design; Design of Experiments; Haptic Systems for Virtual Reality and Teleoperation; Engineering Vibrations; Automatic Feedback Control; Instrumentation and Measurement Systems.

**Campus and Community Involvement**

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|--------------------------------------------------|------------------------|
| Member, American Society of Mechanical Engineers | March 2015 – Present   |
| WPI Video Game Orchestra                         | Nov 2024 – Present     |
| WPI Tennis Club                                  | Aug 2024 – Present     |
| WPI Philharmonic Orchestra, 1st Violinist        | Nov 2023 – Present     |
| CSULB Long Beach Lunabotics, Mechanical Engineer | Sept 2020 – May 2022   |
| CSULB Beach Launch Team, Avionics Engineer       | Sept 2020 – May 2022   |
| CSULB Bob Cole Symphony Orchestra, 2nd Violinist | Sept 2019 – March 2020 |
| Rebuilding Together Solano County, Volunteer     | Aug 2015 – Aug 2016    |